

ETAP Protection Coordination Engineering Report

Industrial 11 kV and 0.415 kV System

Phase and ground fault protection coordination

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System	11 kV utility supply and 0.415 kV LV distribution
Transformer	T1, 2.5 MVA
Major load	Mtr1, 500 kW
Study software	ETAP 19.0.1 Star and Sequence of Operation

Executive summary

This study verifies selective phase and line-to-ground fault protection for an industrial distribution system. The final settings preserve the established phase-coordination study while adding sensitive ground protection. For a Bus5 line-to-ground fault, CB3 ground protection clears in 120 ms and CB2 provides backup in 360 ms. For a Bus4 line-to-ground fault, CB2 ground protection clears in 360 ms, CB2 phase protection provides local backup in 480 ms, and Relay1 phase element 51 provides upstream backup in 3.614 s before CB1 completes opening at 3.634 s.

The selected Relay1 delay is a coordination compromise. A faster setting interfered with the complete phase study, including transformer energization and motor-starting constraints. The retained setting is acceptable for this portfolio model only when its operating point remains below the transformer and conductor damage limits.

1 Study scope

The work builds on an existing ETAP model that already includes load flow, short circuit, device duty, motor starting, phase protection coordination, sequence of operation, and arc-flash analysis. This report focuses on:

- Phase coordination between the motor feeder, LV incomer, and upstream transformer-feeder relay.
- Ground protection selectivity between CB3 and CB2.
- Primary and backup clearing for faults at Bus5 and Bus4.
- Review of AutoStar coordination and cable-protection results.

2 System and assumptions

Item	Study basis
Utility system	11 kV source
Transformer T1	2.5 MVA, supplying the LV system
LV reference voltage	0.415 kV nominal; AutoStar results referenced at 0.426 kV
Motor load	Mtr1, 500 kW
Fault type	Symmetrical RMS line-to-ground fault
Protection chain	CB3 downstream, CB2 upstream LV, Relay1 and CB1 upstream MV
Coordination criterion	Nearest device clears first; upstream device provides delayed backup

2.1 System single line diagram

The protection path runs from the 11 kV utility through CB1 and Relay1, the 2.5 MVA transformer, LV incomer CB2, and the outgoing feeders. CT1 has a 150/1 ratio. The principal studied branch is CB3, Cable3, and the 500 kW motor at Bus5.

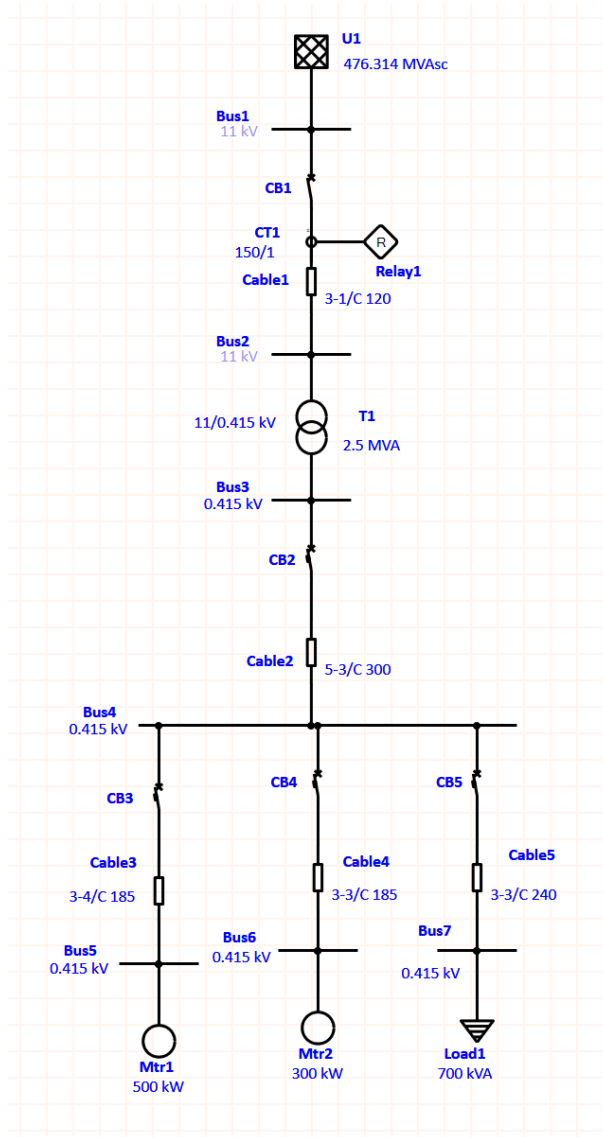


Figure 1 System single-line diagram and protection hierarchy

3 Method

ETAP Star was used to overlay protective-device curves with transformer inrush, transformer withstand, motor-starting, and equipment-damage information. Sequence-of-operation studies were then run for line-to-ground faults at Bus5 and Bus4. AutoStar was used as an independent screening check for coordination and cable protection.

4 Phase protection coordination

The phase study was retained as the governing constraint for the upstream relay. The final arrangement maintains downstream-to-upstream selectivity while allowing motor starting and transformer energization.

Relay1 was not accelerated to the earlier 1.47 s trial setting because that change degraded the complete phase-coordination result.

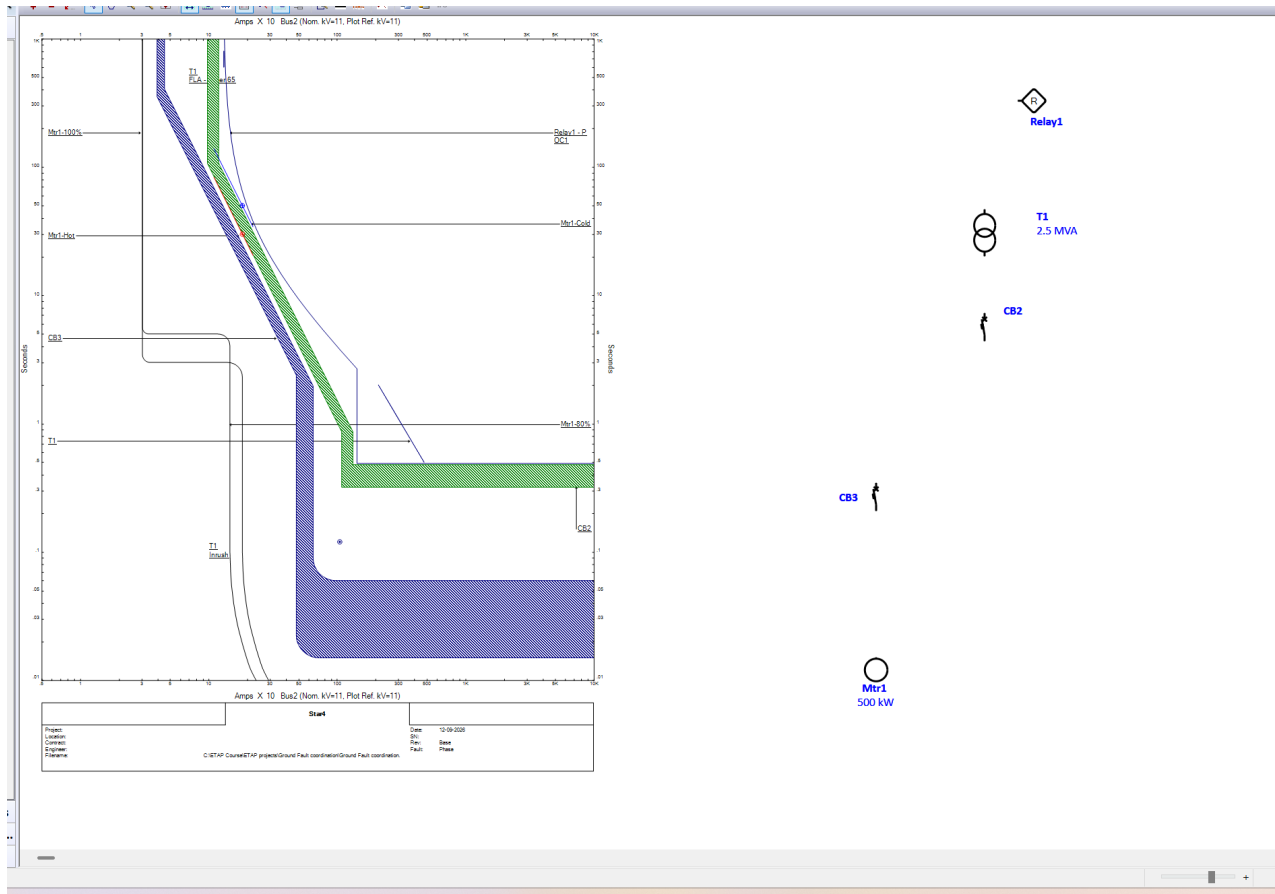


Figure 2 Phase coordination TCC including the 500 kW motor and 2.5 MVA transformer

5 Ground fault coordination

Ground functions were coordinated so that CB3 clears downstream Bus5 faults and CB2 provides delayed backup. At Bus4, CB2 ground protection operates before its phase element. Relay1 responds through phase element 51 for the LV line-to-ground fault; the report does not claim that an MV residual ground element detects LV zero-sequence current through a delta transformer winding.

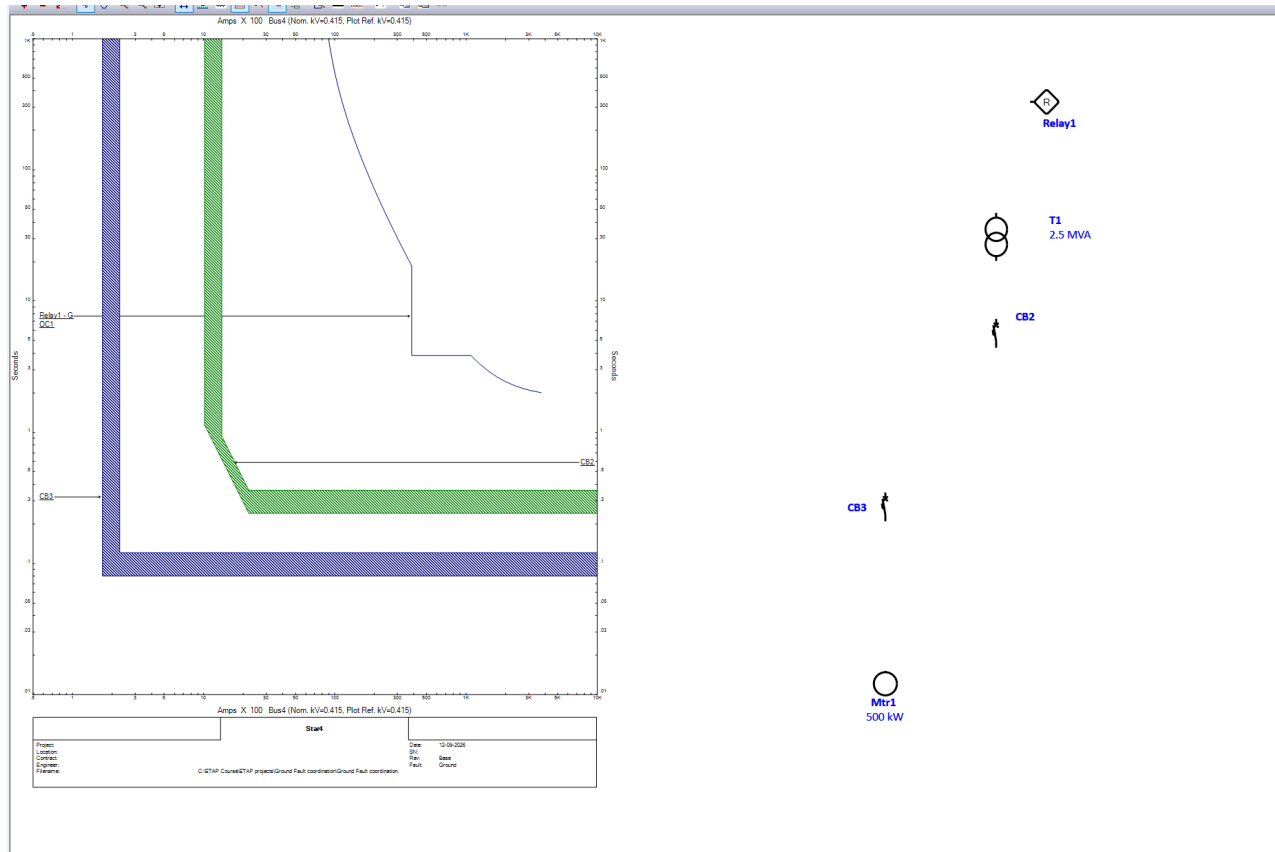


Figure 3 Ground protection coordination TCC

6 Sequence of operation results

Fault	Device and function	T1	Total time	Assessment
Bus5 L-G	CB3 ground	80 ms	120 ms	Primary
Bus5 L-G	CB2 ground	240 ms	360 ms	Backup
Bus4 L-G	CB2 ground	240 ms	360 ms	Primary
Bus4 L-G	CB2 phase	320 ms	480 ms	Local backup
Bus4 L-G	Relay1 phase 51	3.614 s	3.614 s	Upstream backup
Bus4 L-G	CB1 opening	20 ms	3.634 s	Fault isolation

6.1 Bus5 fault

CB3 clears the downstream ground fault in 120 ms. CB2 clears prospectively in 360 ms if CB3 fails. The total-time grading interval is 240 ms. CB2 begins its trip sequence at 240 ms, leaving a 120 ms reset margin after CB3 has completed clearing.

Sequence-of-Operation Events - Output Report: Untitled					
Line-to-Ground (Symmetrical) fault on bus: Bus5					
Data Rev.: Base		Config: Normal		Date: 12-09-2026	
Time ...	ID	If (kA)	T1 (ms)	T2 (ms)	Condition
120	CB3	12.0...	80.0	120	Ground
360	CB2	12.0...	240	360	Ground
4941	CB3	10.8...	3294	4941	Phase
9710	CB2	10.5...	6473	9710	Phase

Figure 4 Bus5 line-to-ground sequence of operation

6.2 Bus4 fault

CB2 ground protection clears the Bus4 fault in 360 ms. CB2 phase protection is slower at 480 ms. Relay1 phase element 51 operates at 3.614 s, followed by the 20 ms CB1 opening time. Later CB3 and CB4 entries shown by ETAP are prospective calculations; CB2 has already isolated the fault.

Sequence-of-Operation Events - Output Report: Untitled					
Line-to-Ground (Symmetrical) fault on bus: Bus4					
Data Rev.: Base		Config: Normal		Date: 12-09-2026	
Time ...	ID	If (kA)	T1 (ms)	T2 (ms)	Condition
360	CB2	55.4...	240	360	Ground
480	CB2	51.0...	320	480	Phase
3614	Relay1	1.093	3614		Phase - OC1 - 51
3634	CB1		20.0		Tripped by Relay1 Phase - OC1 - ...
49487	CB4	1.713	32992	49487	Phase
78345	CB3	2.728	52230	78345	Phase

Figure 5 Bus4 line-to-ground sequence of operation

7 AutoStar evaluation

The exported AutoStar workbook reports the following evaluated pairs as coordinated:

Upstream	Downstream	Fault	Evaluated current range	Status
CB2	CB3	Three phase	1,200 to 54,101 A	Pass
CB2	CB3	Line to ground	230 to 55,492 A	Pass
CB2	CB4	Three phase	960 to 55,817 A	Pass
CB2	CB4	Line to ground	184 to 55,492 A	Pass
CB2	CB5	Three phase	1,392 to 58,763 A	Pass
CB2	CB5	Line to ground	368 to 55,492 A	Pass

8 Observations requiring project data confirmation

AutoStar also produced warnings that should remain visible in the portfolio as engineering follow-up items:

- Bus4 continuous current is zero or its bus type is set to Other, so continuous-current evaluation was not completed.
- Mtr2 has an incomplete or missing motor-starting curve.
- Cable4 phase protection requires review. CB4 long-time pickup is 800 A, slightly above the reported 789.7 A cable ampacity, although the device is the next available setting and the damage curve is protected.
- CB2 long-time pickup is 1,600 A against a reported Cable2 ampacity of 2,682 A. AutoStar labels the cable underutilized but confirms that the damage curve is protected.

9 Engineering checks

Check	Result
CB3 ground clears Bus5 before CB2 ground	Pass
Bus5 ground total-time grading interval	240 ms - Pass for study objective
CB2 ground clears Bus4 before CB2 phase	Pass
Relay1 phase backup remains slower than CB2	Pass
AutoStar evaluated coordination pairs	Pass
Minimum remote-end earth-fault sensitivity	Confirm in final design dataset
CT accuracy and saturation	Not demonstrated
Manufacturer tolerance and breaker selectivity tables	Required before field application
Transformer and conductor withstand at 3.634 s backup clearing	Confirm against final equipment data

10 Conclusion

The final ETAP study demonstrates selective phase and ground-fault protection. Bus5 faults are cleared by CB3 ground protection with CB2 backup. Bus4 faults are cleared by CB2 ground protection with local phase and upstream Relay1 phase backup. The timing preserves the complete phase study and provides a coherent downstream-to-upstream sequence. Field application requires approved equipment, grounding, CT, minimum-fault, and withstand data.